The Amplitude Law: a candidate two-field completion for a horizon-derived MOND scale, and the solar-system obstruction that blocks it

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(Dated: 20 August 2026)

The acceleration scale $a_0 = \kappa c\sqrt{G\rho_\Lambda} = c^2\sqrt{\Lambda/32\pi} = 9.3619\times10^{-11}\,\mathrm{m\,s^{-2}}$ (with $\kappa=\frac{1}{2}$ *fitted*, not derived) reproduces the observed MOND scale from the cosmological constant alone, and the associated interpolation $g_{\rm obs}^2 = g_{\rm bar}^2 + a_0 g_{\rm bar}$ fits the SPARC radial acceleration relation to 0.108 dex. We ask what field theory, if any, makes the corresponding halo density $\rho = \sqrt{GM_b a_0}/(4\pi G r^2)$ a *dynamical consequence* rather than an initial condition. We first show that the equation-of-state route is not the binding constraint: in the weak field the anisotropy $p_r - p_t$ is unobservable (an exact flat direction of the two lensing observables), so the constraint is the amplitude, not the stress. We then exclude several carrier classes outright—every *static* single k-essence scalar obeys $p_t = -\rho$ identically for any kinetic function, and the general khronon fails by a scaling theorem because a_0 cannot be built from c and G alone. Of six candidate mechanisms subjected to adversarial verification, one survives: a two-field lock in which a mediator's Gauss law plus a pointwise balance yields, for *any* free function and with no interpolation invoked, $GM_\chi(r)/r^2 = g_{\rm obs} - g_{\rm bar}$ as an identity—the real dark mass equals the phantom mass. Feeding the interpolation then returns $\rho_\chi = [r/\sqrt{r^2+r_M^2}]\,\sqrt{GM_b a_0}/(4\pi G r^2)$ with the coefficient 0.9999975 and $\mathrm{d}\ln(\mathrm{coef})/\mathrm{d}\ln M_b = 0.500000$, and the screening is keyed structurally on the local total field gradient. It is nonetheless *not* a closure. Its baryon sector is AQUAL as a PDE identity in full generality, so it inherits the modified-gravity solar-system liability: the Cassini quadrupole is 6.7–7.3× the published ceiling, $+18\sigma$ at face value with an honest floor of $\gtrsim 3\sigma$ once the Milky Way mass-model spread is admitted. Two candidate repairs are identified and neither is computed here. We report this as a candidate completion with a live obstruction, not as a completed theory.

I. THE SCALE AND THE QUESTION

The framework studied here takes a single input: that the MOND acceleration scale is set by the cosmological constant through

$$a_0 \;=\; \kappa\, c\sqrt{G\rho_\Lambda} \;=\; c^2\sqrt{\Lambda/32\pi}, \qquad \kappa=\tfrac{1}{2}, \tag{1}$$

giving $a_0 = 9.3619\times10^{-11}\,\mathrm{m\,s^{-2}}$ on the canonical footing ($\rho_{\rm DE}$, cH_Λ) and 1.1279×10^{-10} on an alternative footing ($\rho_{\rm total}$, cH_0). $\kappa=\frac{1}{2}$ **is fitted, not derived**; it is measured at 0.529 ± 0.034, and four candidate values lie within 2σ. Every number in this paper is quoted on both footings.

The framework's own interpolation—not Milgrom's—is

$$g_{\rm obs}^2 = g_{\rm bar}^2 + a_0\, g_{\rm bar} \iff \mu(x) = \frac{\sqrt{1+4x^2}-1}{2x}. \tag{2}$$

Its phenomenology is banked elsewhere: 0.108 dex on the SPARC RAR at $\Upsilon_\star = 0.70$, the baryonic Tully–Fisher relation, and a weak-lensing RAR fit from 40 kpc to 2.2 Mpc.

The question this paper asks is the one that separates a fitting formula from a theory. Equation (2) is equivalent to a halo of density

$$\rho(r) \;=\; \frac{\sqrt{GM_b a_0}}{4\pi G r^2} \tag{3}$$

in the deep regime—isothermal, locked to the *baryonic* mass, with a_0 setting the coefficient. *What makes Eq. (3) a consequence of field equations rather than something put in by hand?* We call this the **amplitude law**, and it is the whole of the closure problem.

II. THE EQUATION OF STATE IS NOT THE CONSTRAINT

It is natural to attack the problem through the halo's stress tensor, because lensing and dynamics respond to different combinations of it. With $\nabla^2\Psi = 4\pi G\rho$ and $\nabla^2\Phi = 4\pi G(\rho + p_r + 2p_t)$, agreement between lensing and dynamics requires $p_r + 2p_t = 0$.

This turns out to be a weak constraint, for two reasons. First, pressureless dust satisfies it trivially. Second, and more sharply, the map $(p_r, p_t) \mapsto (p_r + 2L,\ p_t - L)$ leaves *both* the dynamical and the lensing source unchanged for every L: with three stress unknowns and two observables there is an exact flat direction, and **the anisotropy $p_r - p_t$ is unobservable in the weak field**. The entire testable content is the single number $\epsilon \equiv (p_r + 2p_t)/\rho c^2$.

Calibrating ϵ against data: using the full published 15×15 covariance of the KiDS-1000 excess surface density, the tightest defensible bound is $|\epsilon| < 0.049$, while any non-relativistic carrier has $\epsilon = O(v^2/c^2) \le 1.4\times10^{-4}$. The minimum headroom is 2.53 orders. The equation of state is therefore *not* what constrains this sector; the amplitude is.

III. WHAT IS EXCLUDED

Three classes are closed by theorems rather than by fits.

a. Static single k-essence. For $\mathcal{L} = F(X)$ with a static radial profile $\varphi(r)$ on any static spherically symmetric metric, both $T^t{}_t$ and $T^\theta{}_\theta$ reduce to $-F$, so

$$p_t = -\rho \quad \text{identically, for every } F. \tag{4}$$

The reason is structural: for static φ the gradient reaches the metric only through g^{rr}, so the reduced Lagrangian contains neither g_{tt} nor the areal radius, and the result follows for a *generic* reduced Lagrangian. Shift-symmetric Horndeski G_3 dies with it. Imposing $p_r = -2p_t$ then forces $F \propto X^{3/2}$—the deep-MOND AQUAL power appears unforced from a lensing requirement—but at $p_t/\rho c^2 = -1$, the wrong sign and 6.7 orders from any admissible value.

The escape is the framework's own: a shift-symmetric condensate $\varphi = Q_0 t + \psi(r)$ gives $p_t + \rho = -F' Q_0^2/g_{tt} \neq 0$, and the branch criterion is simply that the condensate flow be subluminal. **The theorem does not touch the framework's dark sector.** It does close every static AQUAL-type scalar as a stress carrier.

b. The radial-director class. Maxwell, general nonlinear electrodynamics, the at-rest khronon, the Einstein-aether vector and a spacelike director were each solved by exact metric variation. Four reach only an order-unity traceless anisotropy; the fifth reaches a small one by being exactly zero. A useful identity emerged: the required tangential pressure is the anomalous field energy density,

$$p_t(r) = \frac{g_{\rm obs}^2 - g_{\rm bar}^2}{8\pi G} = \frac{a_0 g_{\rm bar}}{8\pi G}. \tag{5}$$

c. The khronon, by scaling. The general khronon action contains no dimensionful parameter but G, so its static equations are invariant under $r \to \mu r$, $M \to \mu M$ and any induced density is homogeneous: $\rho_{\rm eff} \sim M^p r^{-2-p}$. The amplitude law needs the mass exponent $\frac{1}{2}$ *and* the radial exponent -2, i.e. $p = \frac{1}{2}$ and $p = 0$ simultaneously. **Incompatible.** Equivalently, a_0 is an acceleration and cannot be built from c and G alone. The bare exterior density is $\propto r^{-4}$, low by 11.7 orders at the MOND radius. The khronon nonetheless *screens* beautifully: the only static scalar available to it is $|\mathrm{d}\ln N|^2 = (g_{\rm total}/c^2)^2$, the total local field gradient, with contrast 3.3×10^7 between the Sun at 1 AU and the Galaxy at the Sun.

IV. THE SURVIVING MECHANISM

Six mechanisms were constructed and each subjected to independent adversarial verification on separate lenses (amplitude, screening, theoretical health). Two were dead on the amplitude law; two fell to their referees—an AQUAL conformal coupling on $c_T \neq 1$, and a mimetic/non-Lagrangian construction on a ghost, $\det Z = -Z_{12}^2 < 0$, removable by no field redefinition. One survives.

The surviving mechanism carries $\Omega_{\rm dm}$ in a dust sector χ and locks it to the baryons through a mediator A with free function μ_v. Its content is: Gauss's law for the mediator, Gauss's law for gravity, and a pointwise balance between them. Combining these eliminates A and gives, **for any μ_v and with no interpolation function used at all**,

$$\boxed{\frac{GM_\chi(r)}{r^2} = g_{\rm obs} - g_{\rm bar}} \tag{6}$$

as an identity. *The real dark mass equals the phantom mass as a theorem, not an assumption.* This is the central result of the paper, and it is kernel-independent.

Feeding Eq. (2) into Eq. (6) gives $M_\chi = M_b(\sqrt{1+u^2} - 1)$ with $u = g_{\rm bar}/a_0$, hence

$$\rho_\chi(r) = \frac{r}{\sqrt{r^2 + r_M^2}} \frac{\sqrt{GM_b a_0}}{4\pi G r^2}, \qquad r_M = \sqrt{GM_b/a_0}. \tag{7}$$

An independent `brentq` plus finite-difference solve on a 200,000-point grid returns log-slope -1.99982, coefficient 0.9999975 of $\sqrt{GM_b a_0}/(4\pi G)$, and $\mathrm{d}\ln(\text{coef})/\mathrm{d}\ln M_b = 0.500000$ over 10^9–$10^{12}\,M_\odot$. A pure-number trap test rejects factors of 2, $\frac{1}{2}$, 4π and $\sqrt{2}$. On a Hernquist source $\rho_\chi > 0$ at 0 of 400,000 radii, so it is not a point-mass artefact. The $1/r^3$ trap that kills k-essence is avoided structurally: μ_v tends to a constant in the deep regime, so the mediator is Maxwell-like there and holds only $\sim 2 \times 10^{-7}$ of the dust's energy.

Screening is structural rather than tuned: μ_v's argument is $|\nabla A|$, and the balance makes $|\nabla A| = |\nabla\Phi|/B$ pointwise, so the free function eats the local *total* field gradient—the one quantity with the required dynamic range. The weak-contrast alternatives (potential 1.51×, dark density 2.22×, dispersion 1.00×) are untouched by the construction.

The mediator's Gauss law caps the dark field at $GM_\chi/r^2 \le C_\nu a_0$ with $C_\nu = 0.5$ exactly for Eq. (2). This is not an extra restriction: $\sup_{g_{\rm bar}}(g_{\rm obs} - g_{\rm bar}) = a_0/2$ exactly for the same interpolation, approached monotonically from below. The cap is the interpolation restating its own supremum. It does *not* forbid clusters: a calibrated cluster ($M_{500} = 5 \times 10^{14} M_\odot$, $R_{500} = 1.3$ Mpc, $f_{\rm bar} = 0.13$, giving $\eta = 1.79$ canonical / 1.64 alt) requires a dark field of $0.383a_0$ / $0.318a_0$, which is 1.30× / 1.57× inside the cap.

V. THE OBSTRUCTION

The mechanism does not close the theory, and the reason is sharp. Applying `sympy` to a general $\Phi(x,y,z)$ with no symmetry assumed gives residual exactly zero for

$$\nabla\cdot\left[\left(1 - \mu_v/B^2\right)\nabla\Phi\right] = 4\pi G\rho_b. \tag{8}$$

The baryon sector is AQUAL as a PDE identity, in full generality. Consequently the statement "baryons carry no dark charge, so there is no fifth force" is true about field content and observationally vacuous: the mechanism inherits the entire modified-gravity solar-system liability, including the external-field-effect quadrupole that constrains the Cassini range.

Evaluated on the operative kernel with the published quadrupole formula, validated first against its own anchors,

$$|Q_2| = 3.46 \times 10^{-26}\,\mathrm{s}^{-2} \text{ (canonical)}, \quad 3.80 \times 10^{-26} \text{ (alt)}, \tag{9}$$

i.e. $6.7\times$ / $7.3\times$ the published 2σ ceiling, or $+18.3\sigma$ / $+20.2\sigma$ at face value. This is corroborated independently: the same interpolation has been published at 8.7σ elsewhere.

Against this finding, and stated because it matters: the published spread of this test is 3–15σ depending on the Milky Way mass model, and +18–20 sits at the top of that range. The honest floor is $\gtrsim 3\sigma$, or $> 1.5\times$ the ceiling at every corner examined. That is a real failure, but not a clean single-number kill.

Separately, Eq. (2) carries a known ephemeris liability: the anomalous sunward acceleration saturates at exactly $a_0/2$, which exceeds the Earth–Mars bound by three to four orders. This liability is *the same algebraic fact* as the theory's realisability: in two-potential form the free function is $\mu_s(s) = k\,s/(1-2s)$ with a simple pole at $s = \frac{1}{2}$, and the pole is what makes Eq. (2) realisable as a coupled scalar at all. The bug and the theory's existence cannot be separated.

VI. WHAT IS NOT CLAIMED

1. **This is not a completed field theory.** It is a candidate completion with one theorem-grade success, one structural success, and one live 3–18σ failure.

2. **The cosmological leg is untested.** Whether the same χ that locks to baryons in a galaxy also delivers $\Omega_{\mathrm{dm}} = 0.265$ to the CMB, without breaking the banked Boltzmann pass, has not been computed.

3. **Ghost-freedom and $c_T = 1$ are uncomputed** for the surviving mechanism's vector sector.

4. **$\kappa = \frac{1}{2}$ remains fitted.** Nothing here derives it, and nothing here constitutes evidence for this interpolation over any other with the same deep limit—the coefficient 1 in the amplitude law is a property of that limit, shared by every MOND kernel.

5. **Clusters remain short by $\approx 2\times$**, inherited unchanged. The mechanism neither creates nor cures the cluster problem.

6. **Two repairs are identified and not computed:** whether a non-static realisation $\varphi = Q_0 t + \psi(r)$ changes the quadrupole, and whether a mediator mass term can suppress the quadrupole-generating region near 5000 AU while leaving galaxies intact. The lane is not closed.

VII. REPRODUCIBILITY

Every load-bearing claim is backed by a committed script that exits zero. The results above were produced under adversarial verification in which independent agents were instructed to refute each claim, with a standing rule that manufacturing a deficit is penalised as heavily as manufacturing a win. Errors were caught in both directions during this work and are logged in the repository, including one case where a first draft asserted an obstruction was fatal at every radius while its own table showed otherwise.

ACKNOWLEDGMENTS

The interpolation function's exponential variant is due to Milgrom & Sanders (2008); the phantom-density formulation is Bekenstein & Milgrom (1984); the aether-scalar-tensor structure invoked as the relativistic setting is due to Skordis & Złośnik. The analysis, scripts and adversarial verification were carried out with Claude (Anthropic).